

1. Database Is a Structured system for storing and organising data so it can be easily accessed.

* Database are the foundation of most modern technology without them, apps, websites and services would fail to deliver personalised and seamless experience.

→ CRMs utilize databases to store Customer Information.

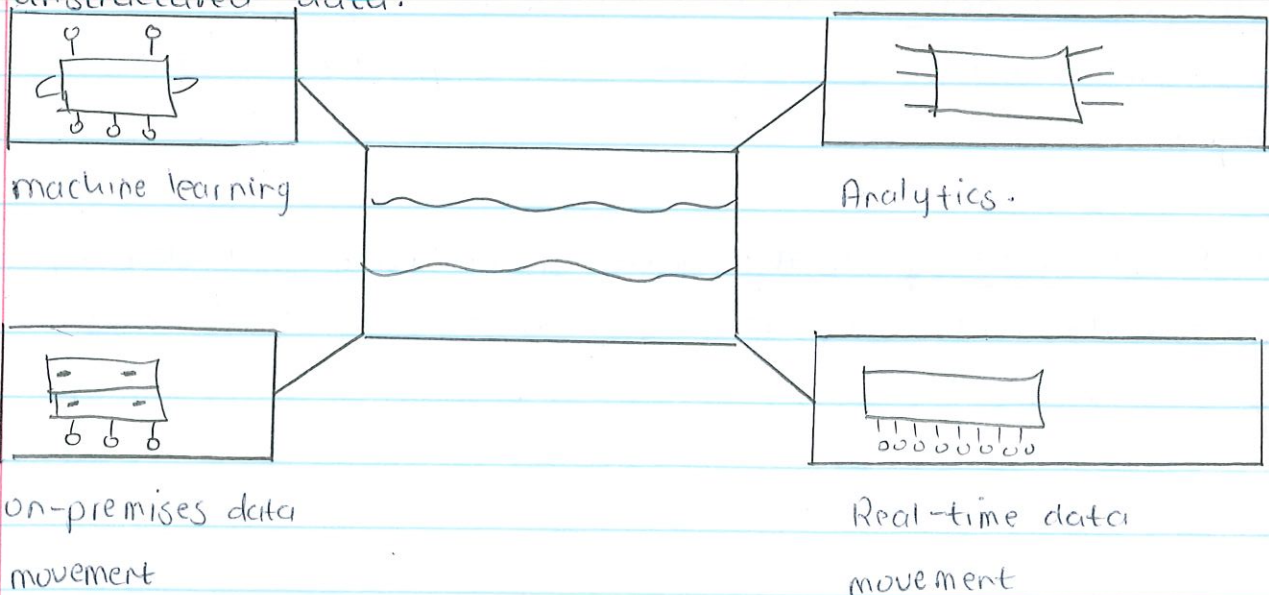
→ Electronic Health Records (EHRs) use data to store and retrieve patient Information..

2. Data Warehouse Is a Centralized repository that stores Structured data (Excell Sheet) and Semi-structured data (webpages)

* A database stores the current data required to power and application whereas a data warehouse stores current and historical data for one or more systems in a predefined and fixed Schema for the purpose of analyzing data.

→ Companies use data warehouse to maintain historical data records.

3. Data lake Is a Centralized repository designed to store, process and Secure large amounts of structured, semistructured and unstructured data.



* Data lake store all types of raw data, which data scientists may use for a variety of projects while data warehouse store cleaned and processed data, which can be used to source analytic or operational reporting.

→ when flexibility, cost efficient storage, and support for diverse or structured data are essential But choose data warehouse when you need fast, reliable access to curated data.

4. Data Mart Is a Subset of a data warehouse focused on a particular line of business, department or Subject area.

* Data marts have a Single-Subject focus and are more decentralised in nature compared to data warehouse which typically store data from multiple business units.

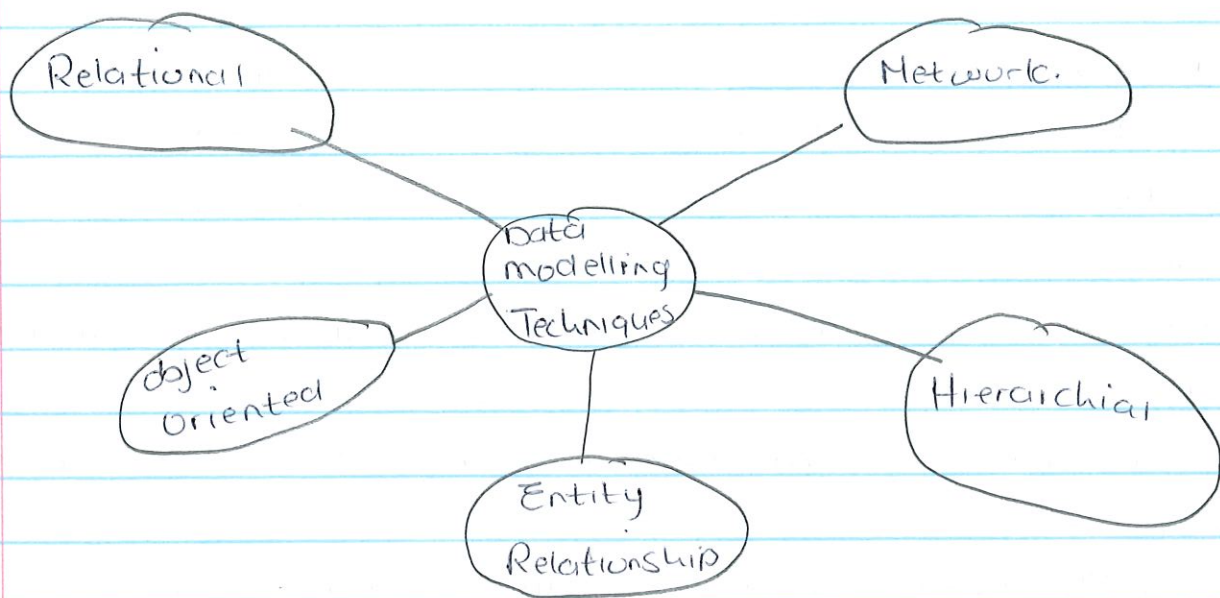
* Marketing team may use data marts to analyze consumer behaviours.

5. Structured data Is a data whose elements are addressed for effective analysis e.g. Relational data.

Semi-Structured data Is information that does not reside in a relational database but has some organizational properties that make it easier to analyze e.g. XML data.

UnStructured data Is data which is not organised in a predefined manner or does not have a predefined data model e.g. Word.

6. Data Modelling Is the process of analyzing and defining all the different data types your business collects and produces.



It is important because you can document what type of data you have, how you use it and the data management requirements surrounding its usage, protection and governance.

→ Conceptual data model - This is a model used in the requirements-gathering process.

→ Representational data model - It is used to represent only the logical part of the database and does not represent the physical structure of database.

→ physical data model - It is used to practically implement Relational data model.

1. Data mining is the process of finding anomalies, patterns and correlations within large data sets to predict outcomes.

* It is used to transform large data sets into actionable business insights.

→ Banks use data mining to better understand market risks.

→ Supermarkets use joint purchasing patterns to identify products.

8. Database management is a software system that allows users to create, define, manipulate and manage database.

* mySQL is a free, open source relational database management system.

* Oracle - Is a Commercial, multi-model database management system.

* PostgreSQL - Is a free-open source relational database management.

* MongoDB - Is an open source, NoSQL, document-oriented database management system.

9. Extract, transform load is a three-step process that retrieves data, transform and improves it and delivers the consolidated data into a data warehouse. The ETL process consolidates data, enabling businesses to analyse it effectively and deliver - Insights.

10. Use CSV for simple tabular data.

- JSON for web APIs and nested data.

- Parquet for big data analytics and storage efficiency.

- Avro used in big data ecosystems like Hadoop and Apache Kafka.

11. Section B - Artificial Intelligence & Machine Learning.

11. AI is a set of technologies that empowers computers to learn, reason and perform a variety of advanced tasks in ways that used to require human intelligence.

* Narrow AI systems are confined to the tasks they were designed for, while General AI would theoretically be able to think, reason and learn as a human does.